# SRAM LLC

## AI Adoption Playbook

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AIML 950 | Human and Machine Intelligence | Northwestern Kellogg

# SRAM makes the parts that make bikes go, stop, and shift

Powering bikes from weekend rides to the Tour de France.

| REVENUE | BRANDS | MTB SHARE |
|---|---|---|
| **$1B+** | **6** | **#1** |
| Private, Chicago-founded | Integrated portfolio | Market leader globally |

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| SRAM | RockShox | Zipp | Quarq | Hammerhead | TIME |
|---|---|---|---|---|---|
| **Drivetrains + Braking** | **Suspension** | **Wheels + Cockpit** | **Power Meters** | **GPS Bike Computer** | **Pedals** |
| Gears, chains, and disc brakes for every bike | Forks and shocks that absorb trail impacts | Aero wheels, handlebars, and stems | Sensors that measure rider output in watts | On-bike computer for navigation and training data | Clip-in pedal systems for road and mountain |

**AXS Wireless Ecosystem | All 6 brands connect wirelessly and share data**

# Going from Early wins to AI-first enterprise

Each step builds the infrastructure and organizational trust the next step requires.

**Step 1**

## Democratize AI Tooling

Month 0-12

GitHub Copilot + Claude Code for engineers

Microsoft CoPilot for business function staff

Quarterly AI innovation forums

**$300K annual cost**
**~$1mm annual cost savings**

**Step 2**

## Enterprise Data Consolidation

Month 0-24

AWS-native data lakehouse on existing AWS infrastructure

SAP ERP + Salesforce CRM connected to central platform

Prerequisite for Steps 3B + 4

**$10mm investment**
**Unlocks significant opportunity downstream**

**Step 3**

## Deploy AI to Operations

Month 0-36

3A: AI support agent for AXS + Hammerhead dealers

3B: SageMaker demand forecasting on pilot SKUs

Bounded, measurable pilots

**3A: ~$100k net ongoing annual savings; revenue uplift due to improved customer retention**
**3B: ~$4.1mm net cost savings annually**

**Step 4**

## Revenue + Innovation

Month 36-60+

AXS Intelligence Platform (unified rider data layer)

OEM Automation + Upgrade Recs
Generative Design + R&D

**~$20M ARR new subscription revenue**
**~$8.5mm annually in revenue uplift at scale**
**~$4mm cost savings annually**

Step 2 runs in parallel with Steps 1 and 3A | Steps 3B and 4 depend on Step 2 data infrastructure

# Building AI fluency from the ground up is the prerequisite

Deploy AI into existing tools, build organizational comfort, and identify champions before committing to Steps 3 and 4.

## GitHub Copilot + Claude Code

**SRAM's software engineers**

AI pair programming in existing IDEs.
Accelerates the technical work Steps 2-4 require.
Integrates directly into existing GitHub workflows.

Annual cost: $150k/yr

**75% of engineers active weekly by Month 6**

## Microsoft CoPilot

**Business function staff**

AI-assisted notes, email, and meeting summarization across all business functions.
Integrates directly into Microsoft 365.

Annual cost: $150k/yr

**60% of staff active weekly by Month 9**

Quarterly AI Innovation Forums | Surface grassroots use cases, identify AI champions

**5+ champions named by Month 12**

**Total investment: ~$300k/yr | Total benefit: ~$1.0mm/yr labor-equivalent | Net value: ~$700k/yr**

STEP 2: ENTERPRISE DATA CONSOLIDATION

# Data infrastructure is the prerequisite, not an optional investment

SRAM has no centralized ERP, CRM, or data warehouse today. Customer data lives in Shopify. AXS telemetry is siloed from Hammerhead ride data.

### SOURCE SYSTEMS

- SAP ERP (order + inventory)
- Salesforce CRM (dealers + warranty)
- AXS + Hammerhead telemetry
- Shopify (DTC orders)
- Quarq power data

### AWS DATA LAKEHOUSE

- Unified SKU-level order data
- Dealer sell-through rates
- PowerBI + Databricks (BI layer)
- Data quality gates enforced

### AI / ML LAYER

- Amazon Bedrock (LLM agents)
- Amazon SageMaker (forecasting)
- Amazon Kendra (doc retrieval)
- Salesforce AgentForce
- All Step 3 + 4 tools run here

24-MONTH INVESTMENT

**$10mm**

SAP + Salesforce + Lakehouse

DOWNSTREAM VALUE

**$36.6mm/yr**

Steps 3B + 4 depend on this

DATA READINESS TODAY

**60-70%**

AXS + Hammerhead pilot-ready

# 12 AI initiatives across four business functions

AI opportunities exist across every function at SRAM.

Quick Win → Medium Effort → Long-Term Bet

## ENGINEERING

### Generative Design

Physical testing is slow and expensive

### AXS Intelligence

Telemetry data is collected but unused

### AI-Tuned Components

Suspension and shifting are static

WHERE AI ADDS VALUE

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| SALES | **OEM Proposal Automation**<br>Proposal cycles are slow, manual | **Part Recommendations**<br>No guided upsell at point of service | **DTC Personalization**<br>Velocio has no targeting capability |

WHERE AI ADDS VALUE

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| SUPPLY CHAIN | **Demand Forecasting**<br>No forecasting system exists today | **Inventory Allocation**<br>Stockouts and rush shipping losses | **Customer Analytics**<br>Shopify data is siloed, unused |

WHERE AI ADDS VALUE

# 12 AI initiatives across four business functions

Start where domain knowledge is high and outcomes are measurable. Support is the proving ground for everything that follows.

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| SUPPORT | RECOMMENDED PILOT<br>**AI Support Agent**<br>70% of dealer questions are repetitive | **Warranty Automation**<br>Fraudulent and mis-filed claims cost time | **Compatibility Assistant**<br>Riders get wrong parts, return them |

START HERE

STEP 3A: THE PROBLEM

# Poor support drives customer churn, not product quality

Riders **love** SRAM hardware. They struggle with firmware updates, part compatibility, and warranty resolution.

TRUSTPILOT RATING

**1.6 / 5.0**

Warranty friction and support delays

AUTOMATABLE

**~70%**

Questions follow documented patterns

TOP COMPLAINTS

**Firmware + Pairing**

Reddit, app stores, Trustpilot

## TODAY

**1. Dealer emails a question**

"Which AXS derailleur firmware works with my 2024 Eagle cassette?"

**2. Agent searches manually**

Opens a 200-page compatibility PDF. Searches wikis and past tickets.

**3. Agent writes a response from scratch**

Types the answer, double-checks part numbers, hopes nothing changed.

**4. Dealer waits ~4 hours for first response**

70% of questions follow the same documented patterns every time.

## WITH AI PILOT

**1. Same question arrives**

Dealer submits via Zendesk. Nothing changes for them.

**2. AI searches approved docs instantly**

Amazon Kendra indexes compatibility tables and firmware notes.

**3. AI drafts a response for the agent**

Amazon Bedrock generates an answer. Agent reviews before sending.

**4. Dealer gets a faster, verified answer**

Target first response: ~2.5 hours. Human approves every response.

# Four roles, three systems, one bounded scope

AXS and Hammerhead dealer support tickets only. Every AI-generated response requires human approval before it reaches a dealer.

## WHO DOES WHAT

### Product Owner
From the support organization

Owns pilot scope and success criteria.
Runs the weekly quality review.
Decides whether to expand, hold, or kill.
Reports to the C-suite AI Sponsor.

### Integration Engineer
1 engineer (internal or contractor)

Connects Zendesk to Amazon Kendra and Bedrock.
Indexes AXS + Hammerhead docs into Kendra.
Maintains the knowledge base as docs update.
Tunes Bedrock prompts based on agent feedback.

### Support Agents
Existing AXS + Hammerhead team

Review every AI-drafted response before sending.
Edit, approve, or reject each draft.
Flag quality issues for weekly review.
Full authority to override any AI output.

### Support Manager
Existing role, expanded scope

Tracks all five SMART goals weekly.
Runs Day-45 and Day-90 gate reviews.
Escalates kill-criteria violations to the PO.
Owns the rollback decision if quality drops.

## WITH WHAT SYSTEMS

### Zendesk
TICKET INTAKE

Receives dealer support tickets.
Routes AXS + Hammerhead tickets to the AI workflow.
All other tickets follow the existing manual process.
No changes to the dealer-facing experience.

### Amazon Kendra
KNOWLEDGE BASE

Indexes SRAM compatibility tables, firmware release notes, and service bulletins.
Searches approved documentation only.
Updated by integration engineer as docs change.

### Amazon Bedrock
RESPONSE GENERATION

Generates draft responses from Kendra search results.
Cites the source document for every claim.
Flags low-confidence answers for manual handling.
Never sends a response without agent approval.

**SCOPE**
AXS + Hammerhead tickets only

**GUARDRAIL**
Human approves every response

**KILL SWITCH**
2 weeks quality drop = rollback

STEP 3A: WHAT WE MEASURE

# Five SMART goals define success at the Day-120 go/no-go gate

Weber set the bar: a 30% reduction in handle time on AXS support tickets with dealer satisfaction scores holding flat or improving.

| GOAL | METRIC | TARGET | OWNER | HOW IT'S MEASURED | TRIGGER IF OFF TRACK |
|---|---|---|---|---|---|
| G1 | Cut first-response time | -40% | Support Manager | Measured weekly vs. current SLA baseline in Zendesk | Below -20% at Day 90 triggers scope review |
| G2 | Resolve without escalation | +15 pts | Support Manager | First-contact resolution rate tracked weekly vs. baseline | Flat or declining at Day 90 triggers agent retraining |
| G3 | AI draft acceptance rate | >70% | Product Owner | % of AI drafts approved by agents without major edits | Below 50% at Day 90 triggers knowledge base rebuild |
| G4 | Agent throughput increase | +25% | Support Manager | Tickets handled per agent per day, tracked weekly | No improvement by Day 90 triggers workflow audit |
| G5 | Hold customer satisfaction flat | No decline | Product Owner | Dealer CSAT survey scores, measured biweekly | Any sustained 2-week drop triggers immediate rollback |

**KILL CRITERIA**

**2 consecutive weeks of quality decline | Any safety error reaching a dealer | Dealer opt-out rate above 15% | Any of these triggers full rollback**

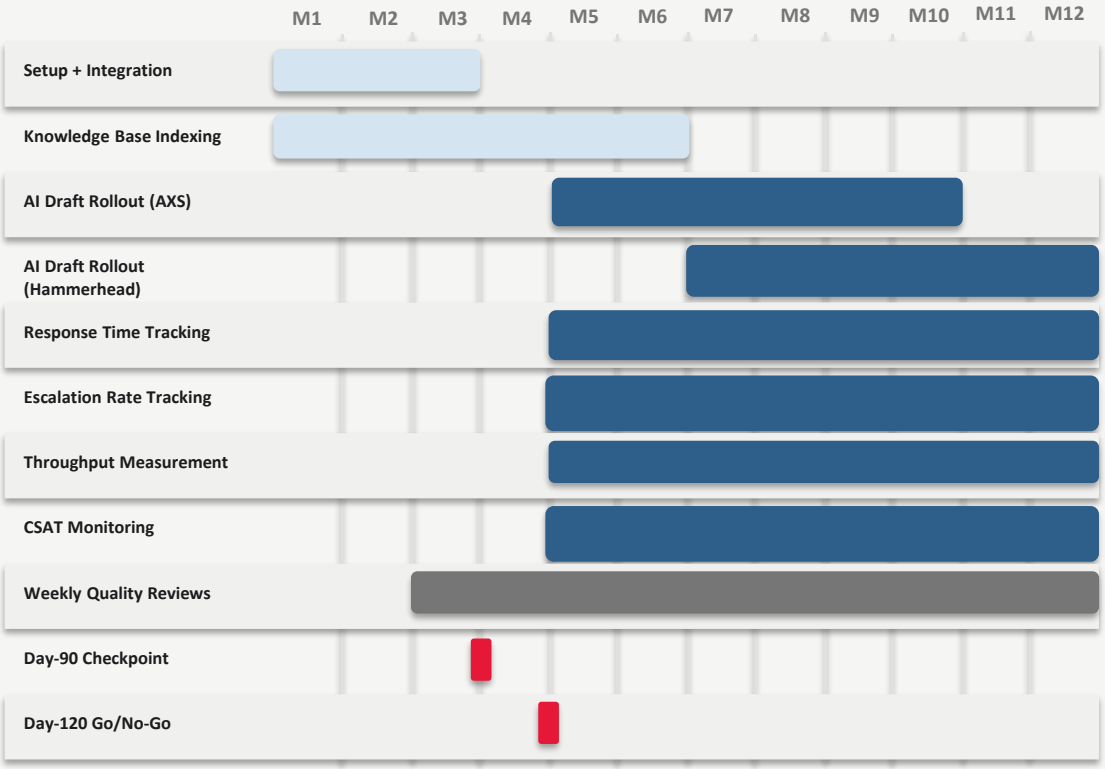
# 12-month pilot

Day-90 scope review, Day-120 go/no-go gate, Month 7-12 scale to full catalog.

## Financial Breakdown

| | |
|---|---|
| **SETUP COSTS** | |
| Zendesk AI + Kendra + Bedrock integration | **$150k** |
| Knowledge base indexing and tuning | **$100k** |
| Integration engineer (6 months) | **$150k** |
| Product owner allocation (6 months) | **$100k** |
| **ANNUAL OPERATING COSTS** | |
| AI platform licensing (Bedrock + Kendra) | **$200k/yr** |
| Knowledge base maintenance | **$50k/yr** |
| Quality review and oversight | **$50k/yr** |
| **ANNUAL GROSS BENEFIT** | |
| Warranty claim accuracy | **$400k/yr** |
| Reduced escalations and handle time | **N/Q savings** |
| Customer retention improvement | **Incremental** |

## 12-Month Timeline



# No forecasting system exists today; SageMaker fills that gap

Stockouts, markdown losses, and overproduction are addressable without touching the product. Requires the Step 2 data foundation as a hard gate.

## TODAY

### 1. Brand-level judgment drives production

No centralized inventory visibility.
Dealer sell-through not systematically tracked.

### 2. Stockouts surface as surprises

Rush shipping absorbs margin.
Markdowns clear overstock at loss.

### 3. Planners react to problems

Decisions made without model support.
Data exists at brand level, not enterprise level.

## WITH SAGEMAKER PILOT

### Month 25-27: Model development

SageMaker trained on unified lakehouse data.
Back-test validates MAPE target of 15% or less.

### Month 27-33: Shadow mode

Model runs in parallel. Planners receive recommendations but make decisions as normal.

### Month 30-36: Active pilot

Planners act on model recommendations with full override authority. Overrides logged.

# Demand forecasting delivers once data infrastructure is live

Begins at Month 25 after the Step 2 data lakehouse is operational. Shadow mode validates accuracy before planners act on recommendations.

| SETUP INVESTMENT | ANNUAL GROSS BENEFIT | NET ANNUAL VALUE | ONE-TIME BONUS |
|---|---|---|---|
| ~$500k | $4.8mm | ~$4.1mm | WC release |
| SageMaker + data science lead | Stockouts + markdowns reduction | After $700K operating cost | |

## Success Criteria

| | | |
|---|---|---|
| **Forecast accuracy (MAPE)** | 15% or less | Back-tested before launch |
| **Stockout reduction** | -30% | Measured against 12-month baseline |
| **Markdown reduction** | -25% | Tracked at SKU level |
| **Planner adoption** | >60% | Recommendations reviewed weekly |

## Timeline

**Month 0-24** — **Data foundation (Step 2)**

SAP, Salesforce, and lakehouse build.
Hard gate for forecasting launch.

**Month 25-27** — **Model development**

SageMaker trained on unified data.
Back-test validates accuracy.

**Month 27-33** — **Shadow mode**

Planners compare model vs. judgment.
No operational changes yet.

**Month 30-36** — **Active pilot**

Live recommendations with override.
Overrides logged for model retraining.

WHAT COULD BE

# Hardware company to performance intelligence by 2031

Each capability creates a new recurring revenue stream. SRAM's installed base of connected components becomes a platform, not a one-time sale.

THE UNLOCK

## AXS Intelligence Platform

**NEW REVENUE: Premium subscription tier**

Unified rider dashboard: power, gearing, suspension, heart rate in one view. No competitor has the product breadth to build this.

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### Predictive Maintenance

Connected components predict chain and brake pad replacement before failure. Installed base becomes recurring revenue.

**NEW REVENUE: Per-device monitoring subscription**

### AI-Tuned Performance

Suspension auto-adjusts to terrain. Shifting optimizes for rider power. Products improve continuously after purchase.

**NEW REVENUE: Premium optimization tier**

### Dealer Intelligence

Telemetry tells dealers which parts approach end-of-life in their area. Proactive ordering replaces reactive.

**NEW REVENUE: Dealer analytics license**

**Hardware sells data access → Data improves performance → Performance sells hardware**

STEP 4: NEW REVENUE STREAMS

# Premium AXS, subscription model, and four distinct revenue paths

Each initiative is viable only after Step 2 data infrastructure is live. Combined, they represent $39M+/yr at maturity.

### AXS Intelligence Platform

Unified rider data: AXS drivetrain, RockShox, Quarq power, Hammerhead GPS in one view. No competitor has the breadth to build this.

$5mm build **~$20mm ARR at scale**

### Generative Design + R&D

AI-assisted design, digital twin simulation, automated patent search. Field telemetry trains AI on actual performance data at scale.

$4mm build **$6mm/yr benefit**

### OEM Proposal Automation

Salesforce AgentForce + CPQ accelerates spec proposals and automates documentation. 1% OEM win rate improvement = $2.5mm/yr.

$1mm build **~$2.5mm/yr benefit**

### Compatible Upgrade Recs

SageMaker recommendation engine lifts basket Value through dealer and DTC channels. Predictive maintenance drives repeat purchases.

$500k build **$4mm/yr benefit**

**Step 4 total: ~$10.5mm investment | $32.5mm/yr at scale**

# Three risks worth naming before the decision

## Dealer trust is fragile

One confident wrong answer from the AI reaches a dealer, and the support team loses credibility it spent years building. Human-in-the-loop and weekly quality reviews mitigate this, but they also slow throughput gains.

**MITIGATION: Kill criteria trigger rollback at first safety error**

## Data consolidation has no executive sponsor

Weber owns sales and manufacturing but not the CRM, ERP, or inventory systems. Phase 2 requires a unified data layer across those systems. Without a senior sponsor, data integration stalls and the pilot stays isolated.

**MITIGATION: CEO assigns cross-functional data owner in parallel**

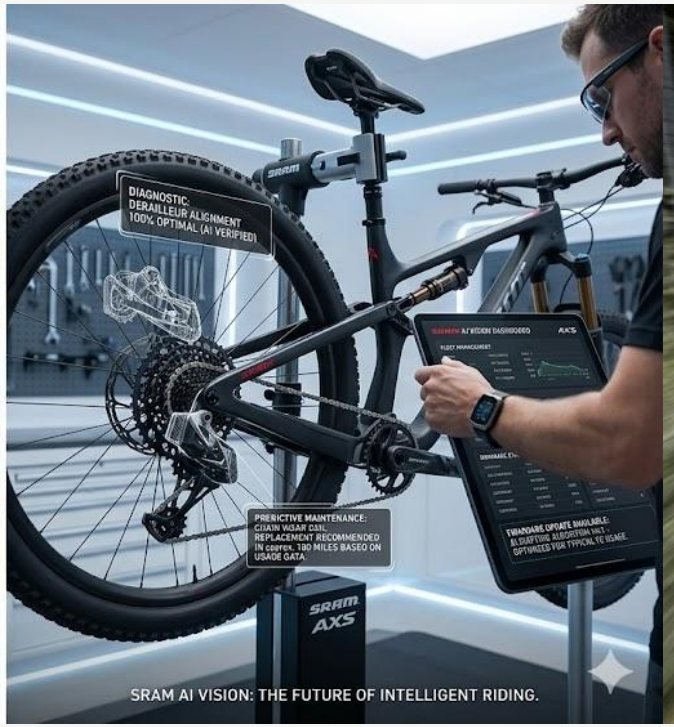
## AI hype creates unrealistic internal expectations

SRAM's engineering culture values precision. If leadership frames AI as a silver bullet rather than a tool, early imperfections will fuel skepticism and internal resistance to future phases.

**MITIGATION: Frame pilot as experiment with defined success and failure criteria**

# Questions

SRAM
AXS
SUSPENSION TERRAIN SCAN
AUTOMATIC DAMPING ADJUSTMENTS
PREDICTIVE SHIFTING
GEAR 10
SHIFT SUGGESTION:
9 - UPCOMING CLIMB PREDICTED
AI COACH
MAINTAIN CADENCE: 85 RPM
HYDRATION REMINDER
SRAM AI VISION: THE FUTURE OF INTELLIGENT RIDING.


DIAGNOSTIC:
DERAILLEUR ALIGNMENT
100% OPTIMAL (AI VERIFIED)
PREDICTIVE MAINTENANCE:
CHAIN WEAR DETECTED
REPLACEMENT RECOMMENDED
IN 200KM. TBD MILES BASED ON
USAGE DATA
SRAM
AXS
SRAM AI VISION: THE FUTURE OF INTELLIGENT RIDING.

SRAM
AXS
SRAM AXS: NEURAL NETWORK ACTIVE
TERRAIN MAPPING & PREDICTIVE SHIFTING [v3.1]
AI-FIRST
BATTERY (AXS): 85%
GEAR: 9th
SHIFT SUGGESTION:
10th (Predicted optimized)
POWER: 220W
HEART RATE: 165 bpm
PREDICTIVE CADENCE: 90 rpm
GEAR: 11th
TIRE PSI: 23
SUSPENSION DATA:
ACTIVE
BATTERY (AXS): 80%
CADENCE: 88 rpm
SUSPENSION DATA:
ACTIVE